Homework 1 - MAE 269

Part 1 - Download and install Python

Part 2 - Python Generation of Kinematics

In [116…
```
import sympy as sp
import numpy as np
from scipy.optimize import minimize
```

In [117…
```
# Defining Symbols
l1,l2 = sp.symbols('l1 l2', real=True, positive=True)
theta1, theta2 = sp.symbols('theta1 theta2', real=True)

# Defining time derivatives
theta1_dot, theta2_dot = sp.symbols('theta1_dot theta2_dot', real=True)
```

In [118…
```
# Function that takes in parameters x, y, & theta and returns a 2D transformation m
def T2D(x, y, theta):
    return sp.Matrix([
        [sp.cos(theta), -sp.sin(theta), x],
        [sp.sin(theta),  sp.cos(theta), y],
        [0,              0,             1]
    ])
```

In [119…
```
# Testing Function
T_test = T2D(1,1,theta1)
T_test
```

Out[119…

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 1 \\ \sin(\theta_1) & \cos(\theta_1) & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

In [120…
```
# Transformation to mass 1
T_0_1 = T2D(l1 * sp.cos(theta1), l1 * sp.sin(theta1), theta1)
T_0_1
```

Out[120…

$$\begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & l_1\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & l_1\sin(\theta_1) \\ 0 & 0 & 1 \end{bmatrix}$$

In [121…
```
# Transformation from mass 1 to mass 2
T_1_2 = T2D(l2 * sp.cos(theta2), l2 * sp.sin(theta2), theta2)
T_1_2
```

Out[121…

$$\begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & l_2\cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & l_2\sin(\theta_2) \\ 0 & 0 & 1 \end{bmatrix}$$

In [122…
```
# Transformation to mass 2
T_0_2 = sp.simplify(T_0_1 * T_1_2)
```

```
T_0_2
```

Out[122…

$$\begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & l_1\cos(\theta_1)+l_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & l_1\sin(\theta_1)+l_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 \end{bmatrix}$$

In [123…

```
# Defining time-dependent joint variables
t = sp.symbols('t')

theta1_t = sp.Function('theta1')(t)
theta2_t = sp.Function('theta2')(t)

# Time-dependent transformation matrices
T_0_1_t = T2D(l1 * sp.cos(theta1_t), l1 * sp.sin(theta1_t), theta1_t)
T_1_2_t = T2D(l2 * sp.cos(theta2_t), l2 * sp.sin(theta2_t), theta2_t)
T_0_2_t = sp.simplify(T_0_1_t * T_1_2_t)
```

In [124…

```
# Defining body twist for mass 1
Vb1_mat = sp.simplify(T_0_1_t.inv() * sp.diff(T_0_1_t, t))
# sp.pprint(Vb1_mat) I don't like the display of sp.pprint it looks funky

Vb1 = sp.Matrix([
    Vb1_mat[1, 0],   # omega_z
    Vb1_mat[0, 2],   # v_x
    Vb1_mat[1, 2]    # v_y
])

Vb1 = sp.simplify(Vb1)
# sp.pprint(Vb1)
```

In [125…

```
Vb1_mat
```

Out[125…

$$\begin{bmatrix} 0 & -\frac{d}{dt}\theta_1(t) & 0 \\ \frac{d}{dt}\theta_1(t) & 0 & l_1\frac{d}{dt}\theta_1(t) \\ 0 & 0 & 0 \end{bmatrix}$$

In [126…

```
Vb1
```

Out[126…

$$\begin{bmatrix} \frac{d}{dt}\theta_1(t) \\ 0 \\ l_1\frac{d}{dt}\theta_1(t) \end{bmatrix}$$

In [127…

```
# Defining body twist for mass 2
Vb2_mat = sp.simplify(T_0_2_t.inv() * sp.diff(T_0_2_t, t))
# sp.pprint(Vb2_mat)

Vb2 = sp.Matrix([
    Vb2_mat[1, 0],   # omega_z
    Vb2_mat[0, 2],   # v_x
    Vb2_mat[1, 2]    # v_y
])
```

```
Vb2 = sp.simplify(Vb2)
# sp.pprint(Vb2)
```

In [128…

```
Vb2_mat
```

Out[128…

$$\begin{bmatrix} 0 & -\frac{d}{dt}\theta_1(t) - \frac{d}{dt}\theta_2(t) & l_1 \sin(\theta_2(t))\frac{d}{dt}\theta_1(t) \\ \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) & 0 & l_1 \cos(\theta_2(t))\frac{d}{dt}\theta_1(t) + l_2\frac{d}{dt}\theta_1(t) + l_2\frac{d}{dt}\theta_2(t) \\ 0 & 0 & 0 \end{bmatrix}$$

In [129…

```
Vb2
```

Out[129…

$$\begin{bmatrix} \frac{d}{dt}\theta_1(t) + \frac{d}{dt}\theta_2(t) \\ l_1 \sin(\theta_2(t))\frac{d}{dt}\theta_1(t) \\ l_1 \cos(\theta_2(t))\frac{d}{dt}\theta_1(t) + l_2\frac{d}{dt}\theta_1(t) + l_2\frac{d}{dt}\theta_2(t) \end{bmatrix}$$

In [130…

```
# Replacing time derivatives with symbolic thetas
subs_dict = {
    sp.diff(theta1_t, t): theta1_dot,
    sp.diff(theta2_t, t): theta2_dot,
    theta1_t: theta1,
    theta2_t: theta2
}

Vb1_sym = sp.simplify(Vb1.subs(subs_dict))
Vb2_sym = sp.simplify(Vb2.subs(subs_dict))

# sp.pprint(Vb1_sym)
# sp.pprint(Vb2_sym)
```

In [131…

```
Vb1_sym
```

Out[131…

$$\begin{bmatrix} \theta_{1dot} \\ 0 \\ l_1\theta_{1dot} \end{bmatrix}$$

In [132…

```
Vb2_sym
```

Out[132…

$$\begin{bmatrix} \theta_{1dot} + \theta_{2dot} \\ l_1\theta_{1dot}\sin(\theta_2) \\ l_1\theta_{1dot}\cos(\theta_2) + l_2\theta_{1dot} + l_2\theta_{2dot} \end{bmatrix}$$

In [133…

```
# Defining Jacobian for mass 1
qdot = sp.Matrix([theta1_dot, theta2_dot])

Jb1 = Vb1_sym.jacobian(qdot)
Jb1 = sp.simplify(Jb1)

# print("Jb1 =")
# sp.pprint(Jb1)

# Defining Jacobian for mass 2
```

```
Jb2 = Vb2_sym.jacobian(qdot)
Jb2 = sp.simplify(Jb2)

# print("Jb2 =")
# sp.pprint(Jb2)
```

In [134…

```
Jb1
```

Out[134…

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ l_1 & 0 \end{bmatrix}$$

In [135…

```
Jb2
```

Out[135…

$$\begin{bmatrix} 1 & 1 \\ l_1 \sin(\theta_2) & 0 \\ l_1 \cos(\theta_2) + l_2 & l_2 \end{bmatrix}$$

In [136…

```
# Defining a function for the world-frame position of mass 2 (end-effector)
p2 = sp.Matrix([
    T_0_2[0, 2], # x_d
    T_0_2[1, 2]  # y_d
])

p2
# sp.pprint(p2)
```

Out[136…

$$\begin{bmatrix} l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \\ l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \end{bmatrix}$$

In [137…

```
# Converting symbolic expression into numerical function
p2_func = sp.lambdify((l1, l2, theta1, theta2), p2, 'numpy')
print(p2_func(1.0, 1.0, np.pi/4, np.pi/4))
```

```
[[0.70710678]
 [1.70710678]]
```

In [138…

```
# Inverse kinematics in order for the end-effector to reach some desired point (x,y

# Desired target
xd, yd = 1.2, 0.8

# Link lengths
l1_val = 1.0
l2_val = 1.0

# Inverse Kinematic Function
def ik_objective(q):
    th1, th2 = q
    pos = np.array(p2_func(l1_val, l2_val, th1, th2), dtype=float).flatten()
    error = pos - np.array([xd, yd])
    return np.dot(error, error)


q0 = np.array([0.5, 0.5])
```

```
result = minimize(ik_objective, q0)

print("Optimization success:", result.success)
print("Optimal joint angles:", result.x)

final_pos = np.array(p2_func(l1_val, l2_val, result.x[0], result.x[1]), dtype=float
print("Final end-effector position:", final_pos)
print("Desired position:", [xd, yd])
```

```
Optimization success: True
Optimal joint angles: [-0.17739072  1.53078645]
Final end-effector position: [1.19999962 0.79999957]
Desired position: [1.2, 0.8]
```

Part 3 - Exploring properties of rotations and transformations in Python

In [139…

```
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Skew-Symmetric Matrix Function
def skew(n):
    nx, ny, nz = n
    return np.array([
        [0,   -nz,  ny],
        [nz,   0,  -nx],
        [-ny, nx,   0]
    ])

# Defining random rotation matrix function
def random_rotation():
    # Random axis
    n = np.random.randn(3)
    n = n / np.linalg.norm(n)

    # Random angle between 0 and 2*pi
    theta = np.random.uniform(0, 2 * np.pi)

    # Rodrigues formula
    n_hat = skew(n)
    R = np.eye(3) + np.sin(theta) * n_hat + (1 - np.cos(theta)) * (n_hat @ n_hat)

    return R
```

In [140…

```
# Testing Function
R = random_rotation()
print(R)
print("det(R) =", np.linalg.det(R))
```

```
[[ 0.9872362  -0.1342742  -0.08564533]
 [ 0.12483145  0.98634294 -0.10744635]
 [ 0.09890293  0.09538369  0.9905151 ]]
det(R) = 0.9999999999999997
```

In [141…

```
# Generating 9 random rotation matrices and computing its determinants
rotations = [random_rotation() for _ in range(9)]
```

```
dets = [np.linalg.det(R) for R in rotations]

for i, det_val in enumerate(dets, 1):
    print(f"Rotation {i}: det = {det_val:.6f}")
```

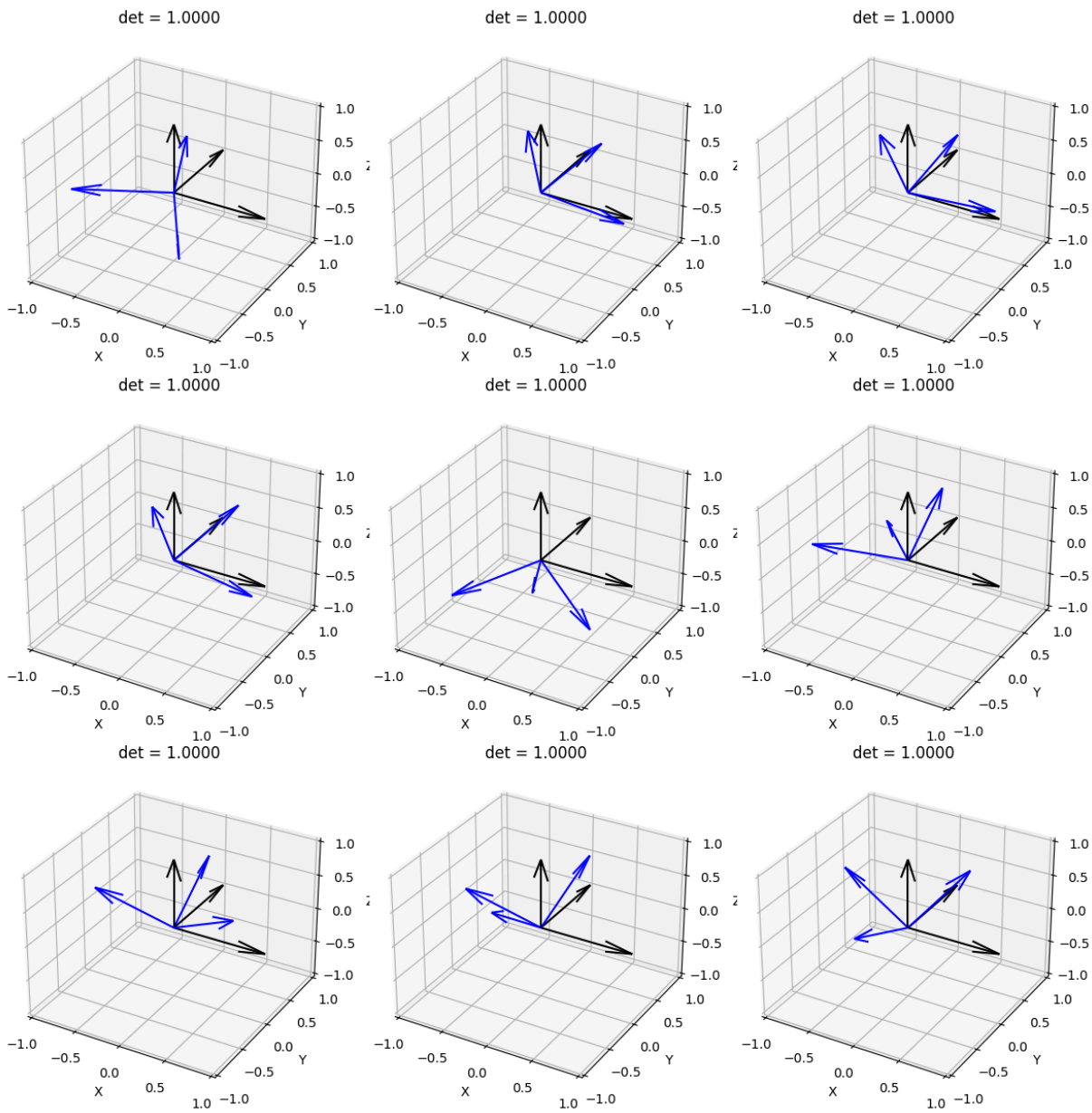
```
Rotation 1: det = 1.000000
Rotation 2: det = 1.000000
Rotation 3: det = 1.000000
Rotation 4: det = 1.000000
Rotation 5: det = 1.000000
Rotation 6: det = 1.000000
Rotation 7: det = 1.000000
Rotation 8: det = 1.000000
Rotation 9: det = 1.000000
```

In [156…

```
# Defining frame plot function
def frame_plot(ax, R):
    origin = np.zeros(3)

    # World frame axes in black
    ax.quiver(*origin, 1, 0, 0, color='k')
    ax.quiver(*origin, 0, 1, 0, color='k')
    ax.quiver(*origin, 0, 0, 1, color='k')

    # Rotated frame axes in blue
    ax.quiver(*origin, *R[:, 0], color='b')
    ax.quiver(*origin, *R[:, 1], color='b')
    ax.quiver(*origin, *R[:, 2], color='b')

    #ax.set_xlim([-1, 1])
    #ax.set_ylim([-1, 1])
    #ax.set_zlim([-1, 1])

    # Use this in order to set the range for each axis you could set an array with
    vals = np.linspace(-1, 1, 5)
    ax.set_xticks(vals)
    ax.set_yticks(vals)
    ax.set_zticks(vals)

    ax.set_xlabel('X')
    ax.set_ylabel('Y')
    ax.set_zlabel('Z')
```

In [157…

```
# Plotting 9 random frames
fig = plt.figure(figsize=(12, 12))

for i in range(9):
    ax = fig.add_subplot(3, 3, i + 1, projection='3d')
    R = random_rotation()
    frame_plot(ax, R)
    ax.set_title(f"det = {np.linalg.det(R):.4f}")

plt.tight_layout()
plt.show()
```



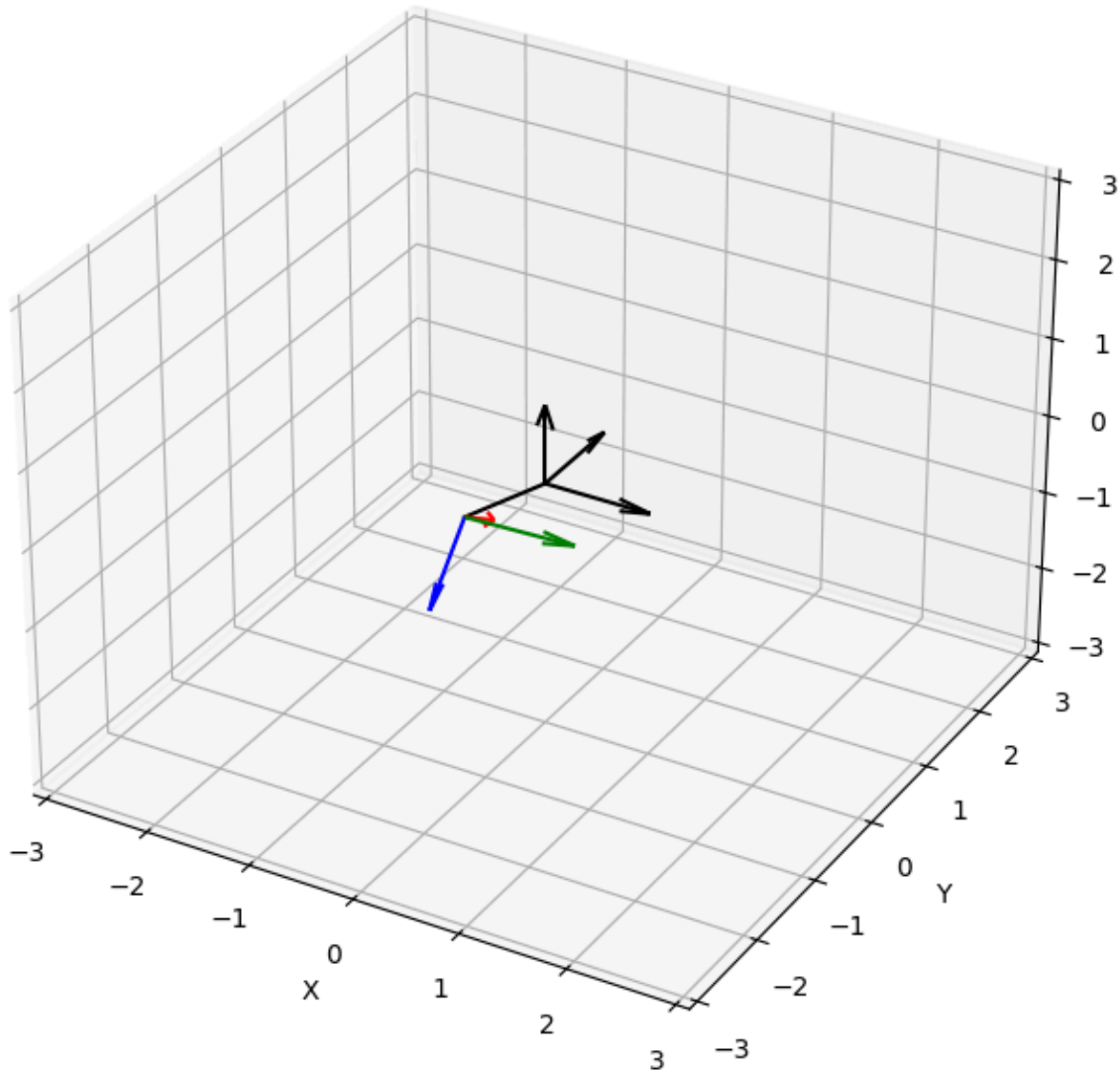
In [158…

```
# Defining function to create random transformations
def random_transform():
    R = random_rotation()
    p = np.random.randn(3)

    T = np.eye(4)
    T[:3, :3] = R
    T[:3, 3] = p
    return T

# Defining function for plotting SE(3) 4x4 transformation matrix (T)
def plot_transform(T):
    fig = plt.figure(figsize=(8, 8))
    ax = fig.add_subplot(111, projection='3d')

    origin = np.zeros(3)
    p = T[:3, 3]
    R = T[:3, :3]
```

```
    # Line from world origin to transform origin
    ax.plot([0, p[0]], [0, p[1]], [0, p[2]], 'k-')

    # World frame
    ax.quiver(0, 0, 0, 1, 0, 0, color='k')
    ax.quiver(0, 0, 0, 0, 1, 0, color='k')
    ax.quiver(0, 0, 0, 0, 0, 1, color='k')

    # Frame at T origin
    ax.quiver(*p, *R[:, 0], color='r')
    ax.quiver(*p, *R[:, 1], color='g')
    ax.quiver(*p, *R[:, 2], color='b')

    ax.set_xlim([-3, 3])
    ax.set_ylim([-3, 3])
    ax.set_zlim([-3, 3])

    ax.set_xlabel('X')
    ax.set_ylabel('Y')
    ax.set_zlabel('Z')
    plt.show()
```

In [145…

```
# Generating and plotting one random transformation matrix
T = random_transform()
plot_transform(T)

print("Translation vector:")
print(T[:3, 3])

print("\nRotation matrix:")
print(T[:3, :3])
```




```
Translation vector:
[-0.76146054 -0.0348727  -0.70453888]

Rotation matrix:
[[-0.1649663   0.98601781  0.0235586 ]
 [ 0.76676128  0.14323425 -0.62574843]
 [-0.62037349 -0.08516358 -0.7796691 ]]
```

In [159…

```
# Generating a random homogeneous vector and verify the length remains the same aft
v = np.random.randn(3) # random 3D vector
R = T[:3, :3]          # transformation

v_rot = R @ v          # rotated vector

# Comparing its lengths
print("Original length:", np.linalg.norm(v))
print("Rotated length: ", np.linalg.norm(v_rot))
```

```
Original length: 0.9738699020384367
Rotated length:  0.9738699020384366
```

Homework 1

Part 3.3 a) $SE(2) \Rightarrow T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix}$, $T^{-1} = \begin{bmatrix} R^T & -R^T t \\ 0 & 1 \end{bmatrix}$ Note $R^{-1} = R^T$ from block inverse formula $A = R, B = t, C = 0, D = 1$

$\Rightarrow A^{-1} = R^T,\ CA^{-1}B = 0,\ D - CA^{-1}B = 1$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{B}\left(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\right)^{-1}\overset{0}{\mathbf{C}\mathbf{A}^{-1}} & -\mathbf{A}^{-1}\mathbf{B}\left(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\right)^{-1} \\ -\left(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\right)^{-1}\underset{0}{\mathbf{C}\mathbf{A}^{-1}} & \left(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}\right)^{-1} \end{bmatrix} = \begin{bmatrix} R^T + \cancel{R^T t}\overset{0}{(1)}(0)R^T & -R^T t(1-0)^{-1} \\ 0 & (1)^{-1} \end{bmatrix} = \begin{bmatrix} R^T & -R^T t \\ 0 & 1 \end{bmatrix} \checkmark$$

$\therefore$ this result matches the geometrically derived inverse T^{-1}, meaning the geometric inverse of a rigid-body transformation is consistent with the standard matrix inverse formula. This only works because rotation matrices are ortnogonal $R^{-1} = R^T$

b) The block matrix of the form: $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ can use the standard block inverse formula only when the block sizes are compatible

A & D must be square matrices, while B & C must must have dimensions that allow the block products to be defined like $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{m \times n}$, $D \in \mathbb{R}^{m \times m}$ with this the overall matrix is square w/ size $(n+m) \times (n+m)$. Additionally, the inverse formula requires A to be invertible and that the matrix $D - CA^{-1}B$ is invertible as well. Otherwise, the block inverse expression is not defined

Part 4

At the beginning of the reading Meyer & Guillot explained how biomimetic realizations are bio-inspired, but bio-inspired are not always biomimetic and I was trying to understand what the difference between the two were until I realized that biomimetic is a subset of bio-inspired. Therefore, bio-inspired is very straightforward ideas or principles that are inspired by biology while biomimetic directly imitates biology (specific structure or behavior). I really liked the idea of the Yale Sonar Robot which can either act as a receiver or a transmitter by using the electrostatic transducers. I thought this was a very interesting robot that could potentially mimic how dolphins are able to locate targets as they are used in the navy in order to find people or robots (enemies) trying to penetrate the perimeter.

One part that was confusing for me was the section about jumping as it does not elaborate on how tangible/beneficial it would be compared to both walking or climbing. It just says it is more efficient in terms of energy but it does not show the disadvantages or other advantages it is kinda vague. Lastly, I really liked the idea of collective robotics where NASA introduced the project of coordinating a fleet of refuelable Entomopter robots I think this could be used in drones where one of the areas of improvement is the duration of the missions need to be short because the batteries do not provide enough power for long distances. Overall, I liked the structure of the reading. It was easy to follow and it built up from one section to another, which made the transition between paragraphs very smooth.